Asymptotic Predictive Risk Bound for the Ridge-Regularised Transfer Learning Estimator in High Dimensions

1 Problem Setup and Notation

Consider a target site t and a source site s. For each site $m \in \{t, s\}$ the response $y^{(m)} \in \mathbb{R}$ and the covariate vector $x \in \mathbb{R}^p$ satisfy

$$y^{(m)} = x^\top \beta^{(m)} + \varepsilon, \qquad x \sim N(0, \Sigma), \quad \varepsilon \sim N(0, \sigma^2),$$

where ε is independent of x and $\beta^{(m)}$. At the target site only the sub-vector $x_{\mathcal{R}(t)}$, indexed by the set $\mathcal{R}(t) \subseteq [p]$ with $|\mathcal{R}(t)| = p_t$, is observed. Write $X \equiv X^{(t)}_{\mathcal{R}(t)} \in \mathbb{R}^{n_t \times p_t}$ for the target design matrix and $Y \equiv Y^{(t)} \in \mathbb{R}^{n_t}$ for the target response.

Given an external estimator $\hat{\beta}^{(s)}$ of $\beta^{(s)}$, define the *projection matrix* $\Phi^{(t)} \in \mathbb{R}^{p_t \times p}$ by

$$\big[\Phi^{(t)}\big]_{\cdot, \mathcal{R}(t)} = I_{p_t}, \qquad \big[\Phi^{(t)}\big]_{\cdot, [\mathcal{R}(t)]^c} = \Sigma^{-1}_{\mathcal{R}(t)} \Sigma_{\mathcal{R}(t), [\mathcal{R}(t)]^c}, \tag{1}$$

and set $\hat{\omega}^{(t)}_s = \Phi^{(t)} \hat{\beta}^{(s)}$. The transfer learning estimator is

$$\hat{\omega}^{(t)}_{\mathrm{TL}} := \operatorname*{arg\,min}_{\omega \in \mathbb{R}^{p_t}} \Big[\frac{1}{n_t} \|Y - X\omega\|_2^2 + \lambda_\omega \|\omega - \hat{\omega}^{(t)}_s\|_2^2 \Big], \qquad \lambda_\omega > 0. \tag{2}$$

For an independent test pair $(\tilde{x}_{\mathcal{R}(t)}, \tilde{y}^{(t)})$ from the target distribution, the *conditional predictive risk* is

$$r^{\mathrm{pred}}\big(\hat{\omega}^{(t)}_{\mathrm{TL}}\big) = \mathbb{E}\Big[\big(\tilde{y}^{(t)} - \tilde{x}^\top_{\mathcal{R}(t)} \hat{\omega}^{(t)}_{\mathrm{TL}}\big)^2 \;\Big|\; X \Big].$$

2 Assumptions

Assumption 2.1 (High-Dimensional Proportional Limit)**.** As $n_t \to \infty$, $p/n_t \to \gamma \in (0, \infty)$ and $p_t/n_t \to \eta \in (0, \infty)$. In particular, $p_t/p \to \eta/\gamma \in (0, 1]$.

Assumption 2.2 (Random Regression Coefficients)**.** For each $m \in \{s, t\}$, $\mathbb{E}[\beta^{(m)}] = 0$ and $\mathrm{Cov}(\beta^{(m)}) = (\alpha_m^2/p)\, I_p$ with $\alpha_m \neq 0$. The cross-covariance satisfies $\mathrm{Cov}(\beta^{(t)}, \beta^{(s)}) = (\rho \alpha_t \alpha_s / p)\, I_p$ with $\rho \in (0, 1]$. Moreover, $\beta^{(t)}$ and $\beta^{(s)}$ are independent of the target design X, the test point $(\tilde{x}, \tilde{y}^{(t)})$, and the target noise.

Assumption 2.3 (Source Estimation Error)**.** The source estimator $\hat{\beta}^{(s)}$ is independent of the target data (X, Y) and the test observation, and satisfies

$$\lambda_{\max}\Big(\mathbb{E}\big[(\hat{\beta}^{(s)} - \beta^{(s)})(\hat{\beta}^{(s)} - \beta^{(s)})^\top \big] \Big) \leq \frac{C^U}{p}$$

uniformly as $(n_t, p) \to \infty$, for a constant $C^U > 0$.

Assumption 2.4 (Population Covariance Regularity)**.** The eigenvalues of Σ are uniformly bounded:

$$0 < c_\ell \le \lambda_{\min}(\Sigma) \le \lambda_{\max}(\Sigma) \le c_u < \infty$$

for constants c_ℓ, c_u independent of p. In particular, $\Sigma_{\mathcal{R}(t)}$ inherits the same bounds. The empirical spectral distribution of $\Sigma_{\mathcal{R}(t)}$,

$$H_{p_t}(\cdot) = \frac{1}{p_t}\sum_{j=1}^{p_t} \delta_{\tau_j}(\cdot), \qquad \tau_1 \ge \cdots \ge \tau_{p_t} > 0 \text{ eigenvalues of } \Sigma_{\mathcal{R}(t)},$$

converges weakly to a compactly supported probability measure H on $(0, \infty)$.

Assumption 2.5 (Bounded Projection Operator)**.** The matrix $\Gamma := \Phi^{(t)}(\Phi^{(t)})^\top$ satisfies $\lambda_{\max}(\Gamma) \le c_\Phi$ uniformly, for a constant $c_\Phi > 0$.

Remark 1. *Assumption 2.5 is implied by Assumption 2.4. Indeed,* $\Phi^{(t)} = \Sigma_{\mathcal{R}(t)}^{-1}\Sigma_{\mathcal{R}(t),[p]}$ *where* $\Sigma_{\mathcal{R}(t),[p]}$ *is the* $p_t \times p$ *sub-matrix of* Σ*. Then* $\Gamma = \Sigma_{\mathcal{R}(t)}^{-1}(\Sigma^2)_{\mathcal{R}(t),\mathcal{R}(t)}\Sigma_{\mathcal{R}(t)}^{-1}$ *and hence* $\lambda_{\max}(\Gamma) \le \|\Sigma_{\mathcal{R}(t)}^{-1}\|_{\mathrm{op}}^2 \|\Sigma\|_{\mathrm{op}}^2 \le c_u^2/c_\ell^2$.

3 Statement of the Main Result

Definition 2 (Marchenko–Pastur companion Stieltjes transform)**.** For $\lambda > 0$, let $v(\lambda) > 0$ be the unique positive solution of the fixed-point equation

$$\frac{1}{v(\lambda)} = \lambda + \eta \int \frac{\tau}{1 + \tau\, v(\lambda)}\, dH(\tau), \tag{3}$$

where H is the limiting spectral distribution of $\Sigma_{\mathcal{R}(t)}$ (Assumption 2.4) and $\eta = \lim p_t/n_t$.

Definition 3 (Asymptotic trace functionals)**.** Define

$$b_0(\lambda) := \int \frac{\tau}{(1 + \tau\, v(\lambda))^2}\, dH(\tau) \cdot \frac{v(\lambda)^2}{1 - \eta \displaystyle\int \frac{\tau^2\, v(\lambda)^2}{(1 + \tau\, v(\lambda))^2}\, dH(\tau)}, \tag{4}$$

$$v_0(\lambda) := \eta \int \frac{\tau}{(1 + \tau\, v(\lambda))^2}\, dH(\tau) \cdot \frac{1}{1 - \eta \displaystyle\int \frac{\tau^2\, v(\lambda)^2}{(1 + \tau\, v(\lambda))^2}\, dH(\tau)}. \tag{5}$$

Theorem 4 (Asymptotic Predictive Risk Upper Bound)**.** *Under Assumptions 2.1–2.5, for every fixed* $\lambda_\omega > 0$ *the conditional predictive risk of the estimator* (2) *satisfies, almost surely as* $(n_t, p) \to \infty$*,*

$$\boxed{\limsup_{n_t, p \to \infty} \mathbb{E}\Big[r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big)\Big] \;\le\; \mathcal{R}_H\big(\lambda_\omega; B_\delta, \sigma_0^2, \eta\big)\,,} \tag{6}$$

where

- $\sigma_0^2 = \dfrac{\alpha_t^2}{p}\operatorname{tr}(\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}) + \sigma^2$ *is the irreducible prediction variance (which converges to a finite limit under Assumption 2.4),*
- $C_\delta = 2\big(h^2 + C^U\big)$ *with* $h^2 = \alpha_t^2 + \alpha_s^2 - 2\rho\,\alpha_t\alpha_s$ *being the transfer discrepancy,*

- $B_\delta = (\eta/\gamma) c_\Phi C_\delta$ *is the effective signal-strength parameter entering the ordinary ridge-risk functional,*
- $c_\Phi \le c_u^2/c_\ell^2$ *is the uniform bound on* $\lambda_{\max}(\Gamma)$*,*
- $\mathcal{R}_H(\lambda; a^2, \tau^2, \eta)$ *is the deterministic equivalent predictive-risk functional for ridge regression with population covariance spectral law* H*, signal strength* a^2*, noise variance* τ^2*, and aspect ratio* η*.*

Corollary 5 (Optimal Regularisation Parameter)**.** *The optimally tuned upper bound can be written without expanding the deterministic-equivalent ridge-risk functional as*

$$\inf_{\lambda_\omega > 0} \limsup_{n_t, p \to \infty} \mathbb{E}\Big[r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) \Big] \;\le\; \inf_{\lambda_\omega > 0} \mathcal{R}_H\big(\lambda_\omega; B_\delta, \sigma_0^2, \eta\big)\,. \tag{7}$$

In the isotropic case $\Sigma = I_p$*, this is minimized at*

$$\lambda_\omega^* = \frac{\gamma\, \sigma_0^2}{c_\Phi C_\delta} = \frac{\gamma\, \sigma_0^2}{2(h^2 + C^U)} \qquad (c_\Phi = 1), \tag{8}$$

and the corresponding unexpanded conclusion is

$$\boxed{\inf_{\lambda_\omega > 0} \limsup_{n_t, p \to \infty} \mathbb{E}\Big[r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) \Big] \;\le\; \mathcal{R}_{\delta_1}\big(\lambda_\omega^*; (\eta/\gamma) C_\delta, \sigma_0^2, \eta\big)\,.} \tag{9}$$

4 Proof of Theorem 4

The proof proceeds in five steps: (i) closed-form solution of the estimator; (ii) oracle target coefficient and noise decomposition; (iii) predictive risk decomposition into bias, variance, and irreducible error; (iv) bounding the bias via the transfer discrepancy; (v) asymptotic evaluation of trace functionals using the Marchenko–Pastur deterministic equivalents of Dobriban and Wager (2018).

4.1 Step 1: Closed-Form Solution

The objective in (2) is strongly convex. Setting the gradient to zero gives

$$\hat{\omega}_{\mathrm{TL}}^{(t)} = \big(\widehat{\Sigma} + \lambda_\omega I_{p_t}\big)^{-1} \Big(\frac{1}{n_t} X^\top Y + \lambda_\omega\, \hat{\omega}_s^{(t)} \Big), \tag{10}$$

where $\widehat{\Sigma} = X^\top X / n_t \in \mathbb{R}^{p_t \times p_t}$ is the sample Gram matrix. We abbreviate $A(\lambda) := (\widehat{\Sigma} + \lambda I_{p_t})^{-1}$ and write $A \equiv A(\lambda_\omega)$.

4.2 Step 2: Oracle Target Coefficient

Because only $x_{\mathcal{R}(t)}$ is observed at the target site, the best (minimum mean-squared-error) linear predictor of $y^{(t)}$ given $x_{\mathcal{R}(t)}$ (conditional on $\beta^{(t)}$) uses the coefficient vector

$$\omega^* := \Sigma_{\mathcal{R}(t)}^{-1} \Sigma_{\mathcal{R}(t),[p]}\, \beta^{(t)} = \Phi^{(t)} \beta^{(t)}, \tag{11}$$

where $\Sigma_{\mathcal{R}(t),[p]} = \mathbb{E}[x_{\mathcal{R}(t)}\, x^\top]$ is the $p_t \times p$ cross-covariance. This follows from the well-known formula for the conditional expectation of a jointly Gaussian vector: $\mathbb{E}[x_{[\mathcal{R}(t)]^c} \mid x_{\mathcal{R}(t)}] = \Sigma_{[\mathcal{R}(t)]^c, \mathcal{R}(t)} \Sigma_{\mathcal{R}(t)}^{-1} x_{\mathcal{R}(t)}$, so that

$$\mathbb{E}[y^{(t)} \mid x_{\mathcal{R}(t)}, \beta^{(t)}] = x_{\mathcal{R}(t)}^\top \big(\beta_{\mathcal{R}(t)}^{(t)} + \Sigma_{\mathcal{R}(t)}^{-1} \Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c}\, \beta_{[\mathcal{R}(t)]^c}^{(t)} \big) = x_{\mathcal{R}(t)}^\top \omega^*.$$

Define the *target residual vector* $\xi \in \mathbb{R}^{n_t}$ entry-wise by

$$\xi_i := y_i^{(t)} - x_{i,\mathcal{R}(t)}^\top \omega^* = \big(x_{i,[\mathcal{R}(t)]^c} - \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)}\Sigma_{\mathcal{R}(t)}^{-1} x_{i,\mathcal{R}(t)}\big)^\top \beta_{[\mathcal{R}(t)]^c}^{(t)} + \varepsilon_i, \qquad i = 1, \dots, n_t. \quad (12)$$

The vector $x_{i,[\mathcal{R}(t)]^c|\mathcal{R}(t)} := x_{i,[\mathcal{R}(t)]^c} - \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)}\Sigma_{\mathcal{R}(t)}^{-1} x_{i,\mathcal{R}(t)}$ is the conditional residual, which by Gaussianity is independent of $x_{i,\mathcal{R}(t)}$ and has covariance $\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)} = \Sigma_{[\mathcal{R}(t)]^c} - \Sigma_{[\mathcal{R}(t)]^c,\mathcal{R}(t)}\Sigma_{\mathcal{R}(t)}^{-1}\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c}$ (the Schur complement). Therefore, conditional on $\beta^{(t)}$, the ξ_i are independent of X with

$$\mathbb{E}[\xi_i \mid \beta^{(t)}] = 0, \qquad \mathrm{Var}[\xi_i \mid \beta^{(t)}] = (\beta_{[\mathcal{R}(t)]^c}^{(t)})^\top \Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}\, \beta_{[\mathcal{R}(t)]^c}^{(t)} + \sigma^2. \quad (13)$$

In matrix form, $Y = X\omega^* + \xi$.

4.3 Step 3: Risk Decomposition

For the test point we write similarly $\tilde{y}^{(t)} = \tilde{x}_{\mathcal{R}(t)}^\top \omega^* + \tilde{\xi}$ with $\tilde{\xi}$ independent of $\tilde{x}_{\mathcal{R}(t)}$ (conditional on $\beta^{(t)}$) and having the same variance (13).

Lemma 6 (Bias–Variance–Noise Decomposition)**.**

$$r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) = \underbrace{\mathbb{E}\big[(\hat{\omega}_{\mathrm{TL}}^{(t)} - \omega^*)^\top \Sigma_{\mathcal{R}(t)}\, (\hat{\omega}_{\mathrm{TL}}^{(t)} - \omega^*) \;\big|\; X\big]}_{=:\, R_{\mathrm{est}}(X)} + \sigma_0^2, \quad (14)$$

where

$$\sigma_0^2 := \mathbb{E}[\tilde{\xi}^2] = \frac{\alpha_t^2}{p}\,\mathrm{tr}(\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}) + \sigma^2. \quad (15)$$

Proof. Since $\tilde{x}_{\mathcal{R}(t)} \sim N(0, \Sigma_{\mathcal{R}(t)})$ is independent of X, Y, and $\tilde{\xi}$, and since $\tilde{\xi}$ is independent of $\tilde{x}_{\mathcal{R}(t)}$ (conditional on $\beta^{(t)}$), the cross-term vanishes:

$$\begin{aligned} r^{\mathrm{pred}} &= \mathbb{E}\big[\big(\tilde{x}_{\mathcal{R}(t)}^\top(\omega^* - \hat{\omega}_{\mathrm{TL}}^{(t)}) + \tilde{\xi}\big)^2 \;\big|\; X\big] \\ &= \mathbb{E}\big[(\omega^* - \hat{\omega}_{\mathrm{TL}}^{(t)})^\top \tilde{x}_{\mathcal{R}(t)}\tilde{x}_{\mathcal{R}(t)}^\top(\omega^* - \hat{\omega}_{\mathrm{TL}}^{(t)}) \;\big|\; X\big] + \mathbb{E}[\tilde{\xi}^2] + 2\,\mathbb{E}\big[\tilde{\xi}\,\tilde{x}_{\mathcal{R}(t)}^\top(\omega^* - \hat{\omega}_{\mathrm{TL}}^{(t)}) \;\big|\; X\big]. \end{aligned}$$

The cross-term is zero because, upon first conditioning on $\beta^{(t)}$ and $\hat{\omega}_{\mathrm{TL}}^{(t)}$, $\mathbb{E}[\tilde{\xi} \mid \beta^{(t)}] = 0$ and $\tilde{x}_{\mathcal{R}(t)}$ is independent of both $\tilde{\xi}$ and $\hat{\omega}_{\mathrm{TL}}^{(t)}$ (given X). Taking the outer expectation over $\tilde{x}_{\mathcal{R}(t)}$ yields $\Sigma_{\mathcal{R}(t)}$.

For (15), use Assumption 2.2: $\mathbb{E}[(\beta_{[\mathcal{R}(t)]^c}^{(t)})^\top \Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}\, \beta_{[\mathcal{R}(t)]^c}^{(t)}] = (\alpha_t^2/p)\mathrm{tr}(\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)})$. □

It remains to bound $R_{\mathrm{est}}(X)$. Substituting $Y = X\omega^* + \xi$ into (10) and using the identity $A\widehat{\Sigma} = I - \lambda_\omega A$ gives

$$\begin{aligned} \hat{\omega}_{\mathrm{TL}}^{(t)} - \omega^* &= A\left(\widehat{\Sigma}\,\omega^* + \frac{X^\top \xi}{n_t} + \lambda_\omega \hat{\omega}_s^{(t)}\right) - \omega^* \\ &= (I - \lambda_\omega A)\omega^* + A\frac{X^\top \xi}{n_t} + \lambda_\omega A\hat{\omega}_s^{(t)} - \omega^* \\ &= -\lambda_\omega A\big(\omega^* - \hat{\omega}_s^{(t)}\big) + A\frac{X^\top \xi}{n_t}. \end{aligned} \quad (16)$$

Define the two components:

$$B_1 := -\lambda_\omega A\big(\omega^* - \hat{\omega}_s^{(t)}\big), \qquad B_2 := A\frac{X^\top \xi}{n_t}. \quad (17)$$

Lemma 7 (Vanishing Cross-Term)**.** $\mathbb{E}[B_1^\top \Sigma_{\mathcal{R}(t)}\, B_2 \mid X] = 0.$

Proof. Condition first on $\beta^{(t)}, \beta^{(s)}, \hat{\beta}^{(s)}$ and X. Then B_1 is measurable with respect to this σ-algebra, while $B_2 = AX^\top \xi / n_t$ has conditional expectation zero because $\mathbb{E}[\xi \mid X, \beta^{(t)}] = 0$ (from (13); the entries ξ_i have zero mean and are independent of X given $\beta^{(t)}$). Since $\mathbb{E}[B_1^\top \Sigma_{\mathcal{R}(t)}\, B_2 \mid X] = \mathbb{E}\big[B_1^\top \Sigma_{\mathcal{R}(t)}\, \mathbb{E}[B_2 \mid X, \beta^{(t)}, \hat{\beta}^{(s)}] \;\big|\; X\big] = 0.$ □

Therefore

$$R_{\text{est}}(X) = \underbrace{\mathbb{E}[B_1^\top \Sigma_{\mathcal{R}(t)}\, B_1 \mid X]}_{\text{Bias}^2} + \underbrace{\mathbb{E}[B_2^\top \Sigma_{\mathcal{R}(t)}\, B_2 \mid X]}_{\text{Variance}}. \tag{18}$$

4.4 Step 4: Bounding the Bias

4a. Transfer discrepancy matrix. Since $\omega^* = \Phi^{(t)}\beta^{(t)}$ and $\hat{\omega}_s^{(t)} = \Phi^{(t)}\hat{\beta}^{(s)}$,

$$\omega^* - \hat{\omega}_s^{(t)} = \Phi^{(t)}\big(\beta^{(t)} - \hat{\beta}^{(s)}\big) =: \Phi^{(t)}\delta_\beta.$$

Lemma 8 (Transfer Discrepancy Bound)**.** *Under Assumptions 2.2 and 2.3,*

$$\mathbb{E}[\delta_\beta\, \delta_\beta^\top] \preceq \frac{C_\delta}{p}\, I_p, \qquad C_\delta := 2\big(h^2 + C^U\big), \quad h^2 := \alpha_t^2 + \alpha_s^2 - 2\rho\, \alpha_t \alpha_s.$$

Proof. Decompose $\delta_\beta = \underbrace{(\beta^{(t)} - \beta^{(s)})}_{=:u} + \underbrace{(\beta^{(s)} - \hat{\beta}^{(s)})}_{=:w}$. By the elementary matrix inequality $(u + w)(u + w)^\top \preceq 2\, uu^\top + 2\, ww^\top$ (which follows from $(u - w)(u - w)^\top \succeq 0$),

$$\mathbb{E}[\delta_\beta \delta_\beta^\top] \preceq 2\, \mathbb{E}[uu^\top] + 2\, \mathbb{E}[ww^\top].$$

From Assumption 2.2,

$$\mathbb{E}[uu^\top] = \mathrm{Cov}(\beta^{(t)}) + \mathrm{Cov}(\beta^{(s)}) - \mathrm{Cov}(\beta^{(t)}, \beta^{(s)}) - \mathrm{Cov}(\beta^{(s)}, \beta^{(t)}) = \frac{\alpha_t^2 + \alpha_s^2 - 2\rho\alpha_t\alpha_s}{p}\, I_p = \frac{h^2}{p}\, I_p.$$

From Assumption 2.3, $\mathbb{E}[ww^\top] \preceq (C^U/p)\, I_p$. Combining yields the result. □

4b. Bias evaluation.

$$\begin{aligned} \text{Bias}^2 &= \lambda_\omega^2\, \mathbb{E}\big[\delta_\beta^\top (\Phi^{(t)})^\top A \Sigma_{\mathcal{R}(t)} A\, \Phi^{(t)} \delta_\beta \;\big|\; X\big] \\ &= \lambda_\omega^2\, \mathrm{tr}\Big(A \Sigma_{\mathcal{R}(t)} A\, \Phi^{(t)} \mathbb{E}[\delta_\beta \delta_\beta^\top] (\Phi^{(t)})^\top\Big) \end{aligned} \tag{19}$$

since δ_β is independent of X (Assumptions 2.2, 2.3). Applying Lemma 8 and $\Gamma = \Phi^{(t)}(\Phi^{(t)})^\top$,

$$\begin{aligned} \text{Bias}^2 &\le \lambda_\omega^2\, \frac{C_\delta}{p}\, \mathrm{tr}\big(A \Sigma_{\mathcal{R}(t)} A\, \Gamma\big) \\ &\le \lambda_\omega^2\, \frac{C_\delta}{p}\, c_\Phi\, \mathrm{tr}\big(A^2 \Sigma_{\mathcal{R}(t)}\big), \end{aligned} \tag{20}$$

where the second inequality uses $\Gamma \preceq c_\Phi I_{p_t}$ (Assumption 2.5) and the cyclic property of the trace ($\mathrm{tr}(A\Sigma_{\mathcal{R}(t)} A\, \Gamma) \le c_\Phi\, \mathrm{tr}(A\Sigma_{\mathcal{R}(t)} A) = c_\Phi\, \mathrm{tr}(A^2 \Sigma_{\mathcal{R}(t)})$).

4.5 Step 5: Variance Computation

Conditional on $\beta^{(t)}$ and X, the entries ξ_i are independent with mean 0 and common variance $\sigma_\xi^2(\beta^{(t)})$ from (13). Therefore

$$\begin{aligned}\mathbb{E}[B_2^\top \Sigma_{\mathcal{R}(t)} B_2 \mid X, \beta^{(t)}] &= \frac{1}{n_t^2}\,\mathbb{E}[\xi^\top X A \Sigma_{\mathcal{R}(t)} A X^\top \xi \mid X, \beta^{(t)}] \\ &= \frac{\sigma_\xi^2(\beta^{(t)})}{n_t^2}\,\mathrm{tr}\big(X A \Sigma_{\mathcal{R}(t)} A X^\top\big) \\ &= \frac{\sigma_\xi^2(\beta^{(t)})}{n_t}\,\mathrm{tr}\big(A \Sigma_{\mathcal{R}(t)} A \widehat{\Sigma}\big), \end{aligned} \tag{21}$$

where we used $\mathbb{E}[\xi\xi^\top \mid X, \beta^{(t)}] = \sigma_\xi^2(\beta^{(t)})\, I_{n_t}$ and the cyclic property. Taking the expectation over $\beta^{(t)}$ (using (15)):

$$\text{Variance} = \frac{\sigma_0^2}{n_t}\,\mathrm{tr}\big(A^2 \Sigma_{\mathcal{R}(t)} \widehat{\Sigma}\big). \tag{22}$$

4.6 Step 6: Asymptotic Evaluation via Marchenko–Pastur Theory

We now appeal to the deterministic-equivalent theory for sample covariance matrices (see Dobriban and Wager, 2018, Theorem 1, and the underlying results of Marchenko and Pastur, 1967; Silverstein and Bai, 1995).

Write $X = Z\Sigma_{\mathcal{R}(t)}^{1/2}$ where $Z \in \mathbb{R}^{n_t \times p_t}$ has i.i.d. $N(0,1)$ entries. Then $\widehat{\Sigma} = \Sigma_{\mathcal{R}(t)}^{1/2}(Z^\top Z/n_t)\Sigma_{\mathcal{R}(t)}^{1/2}$.

Lemma 9 (Deterministic Equivalents)**.** *Under Assumptions 2.1 and 2.4, for each fixed $\lambda > 0$, the following hold almost surely as $n_t \to \infty$:*

$$\frac{1}{p_t}\,\mathrm{tr}\big[(\widehat{\Sigma} + \lambda I)^{-1}\big] \;\longrightarrow\; v(\lambda), \tag{23}$$

$$\frac{1}{p_t}\,\mathrm{tr}\big[(\widehat{\Sigma} + \lambda I)^{-2}\Sigma_{\mathcal{R}(t)}\big] \;\longrightarrow\; b_0(\lambda), \tag{24}$$

$$\frac{1}{n_t}\,\mathrm{tr}\big[(\widehat{\Sigma} + \lambda I)^{-2}\Sigma_{\mathcal{R}(t)}\widehat{\Sigma}\big] \;\longrightarrow\; v_0(\lambda), \tag{25}$$

where $v(\lambda)$ is given by Definition 2, and $b_0(\lambda)$, $v_0(\lambda)$ are given in Definition 3.

Proof. (23) is the classical companion Stieltjes transform convergence for sample covariance matrices with population spectral distribution H and aspect ratio η (Silverstein and Bai, 1995; Dobriban and Wager, 2018, Theorem 1).

For (24), we use the resolvent identity and differentiation. Define $g(\lambda) := \frac{1}{p_t}\mathrm{tr}[(\widehat{\Sigma}+\lambda I)^{-1}\Sigma_{\mathcal{R}(t)}]$. By the deterministic equivalent for general linear functionals of the resolvent (Dobriban and Wager, 2018, Eq. (7)), almost surely

$$g(\lambda) \;\longrightarrow\; \int \frac{\tau}{1 + \tau\, v(\lambda)}\, dH(\tau).$$

Since $\frac{d}{d\lambda}(\widehat{\Sigma} + \lambda I)^{-1} = -(\widehat{\Sigma} + \lambda I)^{-2}$,

$$\frac{1}{p_t}\mathrm{tr}\big[(\widehat{\Sigma} + \lambda I)^{-2}\Sigma_{\mathcal{R}(t)}\big] = -\frac{d}{d\lambda}\, g(\lambda).$$

By the uniform convergence of the Stieltjes transform on compact subsets of $(0,\infty)$ and the fact that the limiting function is analytic, the derivative converges as well. Differentiating the limit:

$$-\frac{d}{d\lambda}\int \frac{\tau}{1+\tau v(\lambda)}\,dH(\tau) = \int \frac{\tau^2 v'(\lambda)}{(1+\tau v(\lambda))^2}\,dH(\tau).$$

To compute $v'(\lambda)$, differentiate the fixed-point equation (3). Writing $F(v,\lambda) := v^{-1} - \lambda - \eta\int \tau/(1+\tau v)\,dH(\tau) = 0$, by the implicit function theorem,

$$v'(\lambda) = -\frac{\partial F/\partial\lambda}{\partial F/\partial v} = -\frac{-1}{-v^{-2} + \eta\int \frac{\tau^2}{(1+\tau v)^2} dH(\tau)} = \frac{-v^2}{1 - \eta v^2 \int \frac{\tau^2}{(1+\tau v)^2} dH(\tau)}.$$

Substituting and simplifying yields (24) with $b_0(\lambda)$ as stated in (4).

For (25), use the algebraic identity $A^2\Sigma_{\mathcal{R}(t)}\widehat{\Sigma} = A\Sigma_{\mathcal{R}(t)} - \lambda A^2\Sigma_{\mathcal{R}(t)}$ (from $A\widehat{\Sigma} = I - \lambda A$, hence $A^2\Sigma_{\mathcal{R}(t)}\widehat{\Sigma} = A^2(\widehat{\Sigma} + \lambda I - \lambda I)\Sigma_{\mathcal{R}(t)} = A\Sigma_{\mathcal{R}(t)} - \lambda A^2\Sigma_{\mathcal{R}(t)}$). Therefore

$$\frac{1}{n_t}\mathrm{tr}(A^2\Sigma_{\mathcal{R}(t)}\widehat{\Sigma}) = \frac{p_t}{n_t}\left[\frac{1}{p_t}\mathrm{tr}(A\Sigma_{\mathcal{R}(t)}) - \lambda\frac{1}{p_t}\mathrm{tr}(A^2\Sigma_{\mathcal{R}(t)})\right] \longrightarrow \eta\left[\int \frac{\tau}{1+\tau v}\,dH(\tau) - \lambda\, b_0(\lambda)\right] =: v_0(\lambda).$$

Substituting the expression for $b_0(\lambda)$ and simplifying yields the formula in (5). □

4.7 Step 7: Assembling the Bound

Combining (14), (18), (20), and (22):

$$r^{\mathrm{pred}} \le \sigma_0^2 + \sigma_0^2 \cdot \frac{1}{n_t}\mathrm{tr}\big(A^2\Sigma_{\mathcal{R}(t)}\widehat{\Sigma}\big) + \lambda_\omega^2 \frac{c_\Phi C_\delta}{p}\mathrm{tr}\big(A^2\Sigma_{\mathcal{R}(t)}\big). \tag{26}$$

Now apply the deterministic equivalents of Lemma 9:

$$\frac{1}{n_t}\mathrm{tr}(A^2\Sigma_{\mathcal{R}(t)}\widehat{\Sigma}) \longrightarrow v_0(\lambda_\omega),$$
$$\frac{1}{p}\mathrm{tr}(A^2\Sigma_{\mathcal{R}(t)}) = \frac{p_t}{p}\cdot\frac{1}{p_t}\mathrm{tr}(A^2\Sigma_{\mathcal{R}(t)}) \longrightarrow \frac{\eta}{\gamma}\, b_0(\lambda_\omega).$$

Substituting into (26) yields the asymptotic bound (6):

$$r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) \;\le\; \sigma_0^2\big(1 + v_0(\lambda_\omega)\big) + \lambda_\omega^2\,\frac{\eta}{\gamma}\,c_\Phi\,C_\delta\,b_0(\lambda_\omega) + o(1) \qquad \text{a.s.}$$

This completes the proof of Theorem 4. □

5 Proof of Corollary 5

Theorem 4 reduces the transfer-learning risk bound to the ordinary deterministic-equivalent ridge-risk functional $\mathcal{R}_H(\lambda_\omega; B_\delta, \sigma_0^2, \eta)$, with $B_\delta = (\eta/\gamma)c_\Phi C_\delta$. Taking the infimum over $\lambda_\omega > 0$ immediately gives (7).

In the isotropic case $H = \delta_1$ and $c_\Phi = 1$. The classical ridge-risk optimizer has the signal-to-noise form $\lambda_\omega^* = \eta\sigma_0^2/B_\delta$, which equals $\gamma\sigma_0^2/C_\delta$ because $B_\delta = (\eta/\gamma)C_\delta$. Substituting the coarse discrepancy constant $C_\delta = 2(h^2 + C^U)$ gives (8), and (9) is just the same optimized bound written without expanding $\mathcal{R}_{\delta_1}$ into its trace formula. □

6 Discussion

Interpretation. The bound (6) decomposes the predictive risk into three interpretable components:

(i) **Irreducible error** σ_0^2: the cost of observing only p_t of the p features, plus the intrinsic noise σ^2. This term vanishes if all features are observed ($p_t = p$, $\Sigma = I$) and $\sigma^2 = 0$.

(ii) **Variance** $\sigma_0^2\, v_0(\lambda_\omega)$: the estimation variance of the ridge estimator, amplified by the effective noise σ_0^2 and modulated by the high-dimensional trace functional v_0. This term *decreases* in λ_ω.

(iii) **Bias** $\lambda_\omega^2(\eta/\gamma)\, c_\Phi C_\delta\, b_0(\lambda_\omega)$: the regularisation bias from shrinking toward the (imperfect) source estimate. This term *increases* in λ_ω. The transfer discrepancy $h^2 = \alpha_t^2 + \alpha_s^2 - 2\rho\alpha_t\alpha_s$ measures the distance between the source and target tasks, and C^U measures the estimation quality of the source.

When does transfer help? Transfer learning improves over the target-only ridge estimator (which corresponds to $\lambda_\omega \to \infty$ or, equivalently, ordinary ridge with no source information) when C_δ is small relative to σ_0^2, i.e., when the source-target discrepancy h^2 and the source estimation error C^U are jointly small compared to the target noise level.

Connection to Dobriban–Wager and Gu–Han–Duan. Our proof strategy follows the general blueprint of Dobriban and Wager (2018) for the asymptotic trace computations (Lemma 9) and the bias-bounding strategy of Gu, Han, and Duan (2025) for the L_2-penalised transfer learning estimator (Steps 3–4, especially the use of the elementary matrix inequality $(u+w)(u+w)^\top \preceq 2uu^\top + 2ww^\top$ and the spectral bound on the projection matrix Γ).

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